

Supplementary Information

LongPhase 2: uncertainty-aware co-phasing of small variants, structural variants and DNA methylation from long reads

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Supplementary Notes

Supplementary Note 1. Reading the phasing graph export

With `-dot`, `longphase phase` writes one file per chromosome, `<prefix>.<chrom>.dot`, containing one line per voting pair, `P.a -> Q.b [label=w]`, where `P` and `Q` are 1-based positions, `a`, `b` $\in \{1, 2\}$ index the reference (1) and alternate (2) allele, and `w` is the weighted haplotype vote accumulated at `P` for that allele at the time the vote was cast. The companion VCF carries, for every phased variant, the fields `INFO/H1`, `INFO/H2` (weighted votes for haplotype 1 and 2) and `INFO/PE` (their normalized Shannon entropy). The `gnn` sub-command reads both files; users who wish to inspect why a variant was unphased can therefore reconstruct its local window without re-running the phaser.

Supplementary Note 2. Reproducing the demonstration

The repository ships a 500-kb demonstration (`chr1:121.5–122.0 Mb`; `demo/demo.sh`) that runs `modcall`, SNV-only and four-class `phase`, `gnn` correction of both outputs and `compare` against a truth VCF, and prints switch errors, phased SNV counts and Hamming distance for the four configurations. Total runtime is under two minutes on four threads.

Supplementary Tables

Supplementary Table 1: Node features of the graph neural network. Each allele node of a window carries 31 features; # is the index in the feature vector. h_1, h_2 : weighted haplotype votes (taken from the DOT edge labels when available, otherwise from INFO/H1, INFO/H2); PE: phasing entropy; PS: phase set; “centre” is the trigger variant of the window. Genomic-context features are computed from the reference when -r is given and are zero otherwise.

#	Feature	Definition
<i>Phasing evidence (16)</i>		
1	PE	normalized Shannon entropy of (h_1, h_2) , 0–1 bit
2	HP	haplotype of this allele node (genotype index)
3	gt_ref	genotype index of the reference allele (0)
4	gt_alt	genotype index of the alternate allele (1)
5	is_center	1 if the variant is the window trigger
6	is_same_ps	1 if the variant shares the trigger’s PS
7	allele	0 for the reference-allele node, 1 for the alternate-allele node
8	rel_offset	$(\text{pos} - \text{pos}_c) / \max \text{pos} - \text{pos}_c $ over the window, signed
9	H1	h_1
10	H2	h_2
11	total_votes	$h_1 + h_2$
12	vote_ratio	$\max(h_1, h_2) / (h_1 + h_2)$
13	log_total	$\log_2(1 + h_1 + h_2)$
14	log_h1	$\log_2(1 + h_1)$
15	log_h2	$\log_2(1 + h_2)$
25	rel_vote_depth	$\log_2((h_1 + h_2) / \overline{(h_1 + h_2)}_{\pm 10 \text{ variants}})$
<i>Genomic context (6)</i>		
16	hp_len	homopolymer run length at the variant (capped at 20)
17	gc_content	GC fraction in ± 50 bp
18	str_context	longest tandem-repeat unit count (unit 2–4 bp) in ± 15 bp, /10, capped at 1
19	seq_entropy	Shannon entropy of the ± 15 bp sequence (bits)
20	neigh_dens	$(\text{number of variants within } \pm 500 \text{ bp} - 1) / 20$
21	dist_near_indel	$\log_{10}(1 + \text{distance to nearest indel})$, capped at $\log_{10}(10001)$
<i>Graph structure (5)</i>		
22	out_degree	number of distinct target positions of outgoing DOT edges, /10, capped at 1
23	ps_block_size	$\log(1 + \text{variants in the PS}) / \log(101)$
24	log_center_dist	$\log_{10}(1 + \text{pos} - \text{pos}_c)$
26	out_weight_std	s.d. of outgoing DOT edge weights, /100, capped at 1
27	is_bridge	1 if removing the variant disconnects the window graph
<i>Variant class (4, one-hot)</i>		
28–31	is_SNV, is_indel, is_SV, is_5mC	variant class indicators

Supplementary Table 2: Edge features of the graph neural network. Edges are taken from the DOT export, restricted to node pairs inside the window, symmetrized, and complemented by one self-loop per node whose features are the mean over the node’s incident edges. w is the DOT edge label (accumulated vote weight).

#	Feature	Definition
1	weight	$1/(1 + e^{-w})$
2	log_dist	$\log_{10}(1 + \text{pos}_{\text{dst}} - \text{pos}_{\text{src}})$
3	log_weight	$\log(1 + w)$
4	is_same_hp	1 if the two allele nodes have the same allele index
5	is_inter_ps	1 if the two variants belong to different phase sets
6	weight_ratio	$w / (\text{sum of outgoing edge weights of the source allele node})$

Supplementary Table 3: Architecture and inference hyperparameters of the graph neural network.

Component	Setting
Input normalization	BatchNorm over 31 node features (frozen statistics at inference)
Input projection	Linear $31 \rightarrow 128$, GELU
Layers	4 GPS layers, hidden size 128
Local branch	GATv2, 4 heads $\times$ 32, LeakyReLU slope 0.2, per-layer edge encoder $6 \rightarrow 128$
Global branch	multi-head self-attention, 4 heads, Q/K/V/O projections $128 \rightarrow 128$
Fusion	$x \leftarrow x + \text{local} + \text{global}$; LayerNorm ($\varepsilon = 10^{-5}$)
Feed-forward	$128 \rightarrow 256 \rightarrow 128$, GELU, residual, LayerNorm
Classifier	$\text{concat}[h_i, \sum_j w_{ij}, \max_j w_{ij}] (130) \rightarrow 128$ (GELU) $\rightarrow 2$, softmax
Parameters	687,358 (float32, 2.75 MB, embedded in the binary)
Window	trigger $\text{PE} \geq 0.80$; ± 20 variants; ≤ 256 nodes
Decision	unphase if mean $P(\text{error}) \geq 0.30$ over the two allele nodes and all covering windows
Post-processing	split disconnected phase sets (largest component keeps PS); bridge protection off
Training	[data, labels, loss, optimizer, epochs]

Supplementary Table 4: Parameters of longphase phase and their defaults.

Parameter	Default	Meaning
-ont / -pb	–	sequencing platform (enables nanopore-specific homopolymer filter)
-indels	off	phase small indels
-indelQuality	0	exclude indels with QUAL below this value
-sv-file, -mod-file	–	SV VCF (Sniffles2/cuteSV with read names); 5mC VCF from modcall
-q, -mappingQuality	1	minimum mapping quality of alignments
-x, -mismatchRate	3	mark reads as unreliable above this mismatch rate
-p, -baseQuality	12	base quality below which an observation is down-weighted
-e, -edgeWeight	0.1	weight of a down-weighted observation (otherwise 1)
-a, -connectAdjacent	35	number of downstream variants each variant is linked to
-d, -distance	300,000	maximum distance (bp) between linked variants
-l, -edgeThreshold	0.7	pair casts no vote if $\min(P, Q) / \max(P, Q)$ exceeds this (0.3 for SNV–5mC)
-L, -overlapThreshold	0.2	overlap fraction above which the shorter of two alignments of a read is dropped
-w, -sv-window	20	CIGAR operations scanned around an SV start
-h, -sv-threshold	0.10	relative length tolerance for a CIGAR indel to support an SV
-m, -readConfidence	0.65	minimum majority fraction to tag a read in read-based correction
-n, -snpConfidence	0.75	minimum confidence to keep a variant’s phase after read-based correction
-dot	off	export the phasing graph (required by gnn)

Supplementary Table 5: Parameters of `longphase modcall` and their defaults. Values are those set in the source; the on-line help text lists 0.7 for `-heterRatio` and 6 for `-connectAdjacent` ([\[reconcile\]](#)).

Parameter	Default	Meaning
<code>-m, -modThreshold</code>	0.8	ML probability at or above which a base is methylated
<code>-u, -unModThreshold</code>	0.2	ML probability at or below which a base is unmethylated
<code>-e, -heterRatio</code>	0.6	minimum $\min(m, u) / \max(m, u)$ for a heterozygous site
<code>-i, -noiseRatio</code>	0.2	maximum fraction of reads with intermediate probability
<code>-a, -connectAdjacent</code>	20	neighbouring variants searched for SNV co-segregation
<code>-c, -connectConfidence</code>	0.9	minimum majority ratio of co-segregating reads
<code>-all</code>	off	also output homozygous modification sites

Supplementary Table 6: Parameters of `longphase gnn` and `longphase haplotag`.

Parameter	Default	Meaning
<i>gnn</i>		
<code>-s, -snp-file</code>	–	phased SNV/indel VCF from <code>phase</code>
<code>-sv-file, -mod-file</code>	–	phased SV and 5mC VCFs (corrected copies are written)
<code>-dot-prefix</code>	from <code>-s</code>	prefix of the DOT files
<code>-r, -reference</code>	–	reference FASTA (enables genomic-context features)
<code>-pe-threshold</code>	0.80	phasing entropy that opens a window
<code>-window</code>	20	variants on each side of the trigger
<code>-B, -break-threshold</code>	0.30	error probability at or above which a variant is unphased
<code>-respect-bridge</code>	off	never unphase bridge vertices
<code>-no-split-blocks</code>	off	keep the original PS even if the block becomes disconnected
<i>haplotag</i>		
<code>-q, -qualityThreshold</code>	1	minimum mapping quality of tagged alignments
<code>-p, -percentageThreshold</code>	0.6	minimum fraction of a read's informative alleles on the majority haplotype
<code>-tagSupplementary</code>	off	also tag supplementary alignments
<code>-sv-file, -mod-file</code>	–	use phased SVs and 5mC sites as additional tagging evidence
<code>-cram</code>	off	write CRAM instead of BAM

Supplementary Table 7: Software and data used in the benchmarks. [Complete versions and accessions.]

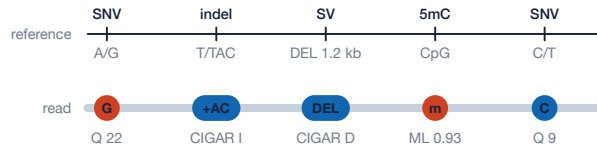
Item	Version	Notes
HG002 nanopore R10.4.1 reads	[–]	[flow cell, kit, basecaller model, source]
Reference	GRCh38 [build]	
minimap2	[–]	-ax map-ont -y
Clair3	[–]	SNV and indel calls
Sniffles2	[–]	-output-rnames
GIAB v4.2.1 HG002	v4.2.1	GRCh38-based small-variant benchmark
HG002 Q100 (T2T-based)	[v?]	referred to as v5.0q
WhatsHap	2.8	[options]
HapCUT2	1.3.4	[options]
LongPhase 1.0	1.0.x [exact]	-ont
LongPhase 2	[release]	-ont; gnn with defaults

Supplementary Table 8: Selected published tools, pipelines and studies that use LongPhase.
Compiled from Europe PMC (80 indexed citing articles) and OpenAlex (96 citing works) for 10.1093/bioinformatics/btac058, September 2026; the “use” column quotes or paraphrases the citing article’s Methods.

Work	Journal, year	Use of LongPhase
<i>Variant callers and pipelines</i>		
ClairS ¹	Nat. Methods 2026	germline phasing and haplotagging of tumour and normal alignments
ClairS-TO ²	Nat. Commun. 2025	phasing of input alignments
Clair3-RNA ³	Nat. Commun. 2025	phasing in the long-read RNA pipeline
Clair3 (accelerated) ⁴	Bioinformatics 2026	intermediate variant phasing and haplotagging
DeepVariant local haplotagging ⁵	Nat. Commun. 2024	cited as haplotagging method used by Clair3/Medaka
Severus ⁶	Nat. Biotechnol. 2025	haplotagged BAMs for somatic SV calling
EPI2ME wf-human-variation ⁷	ONT workflow	<code>-use_longphase</code> phasing
HiFi-somatic-WDL ⁸	PacBio workflow	optional phasing tool
cuteHap ⁹	Adv. Sci. 2026	read phasing before haplotype-aware SV calling
SpecImmune ¹⁰	Adv. Sci. 2026	joint SNV–SV phasing of immune genes
<i>Biological and clinical studies</i>		
Fluctuating DNA methylation in cancer ¹¹	Nature 2025	read phasing (v1.7) and <code>modcall</code> for CpG methylation
Advanced cancer cohort (POG) ¹²	Cell Genomics 2024	discussed as phasing method for tumours
Pediatric epilepsy epesignatures ¹³	Nat. Commun. 2024	phasing of Clair3 and SV calls
GNAS transcripts in PHP1B ¹⁴	PNAS 2025	phasing of Clair3 calls
3-hour genome for neonatal risk ¹⁵	Genet. Med. Open 2024	real-time phasing during sequencing
Ultrarapid diagnostics in critical care ¹⁶	Eur. J. Hum. Genet. 2025	read phasing (v1.0.7)
HYDIN in primary ciliary dyskinesia ¹⁷	Eur. J. Hum. Genet. 2024	phasing of Clair3 calls
Antisense oligonucleotides in APBD ¹⁸	Nucleic Acids Res. 2025	phasing of Clair3 calls
Native CpG methylation fine-mapping ¹⁹	HGG Adv. 2025	phasing and haplotagging (chosen for speed and SV support)
Canadian national platform benchmark ²⁰	Front. Genet. 2026	phasing inside the Clair3 workflow
Pallas’s cat genome ²¹	NAR Genom. Bioinform. 2023	haplotagging for allele-specific methylation
<i>Methods evaluated against LongPhase</i>		
HiPhase ²²	Bioinformatics 2024	comparison method
Ralphi ²³	Genome Res. 2025	comparison method (fastest runtime on ONT/PacBio)
GCphase ²⁴	BMC Bioinformatics 2024	comparison method
Duet ²⁵	BMC Bioinformatics 2022	comparison method for SV phasing

Supplementary Figures

a Allele observations from one alignment



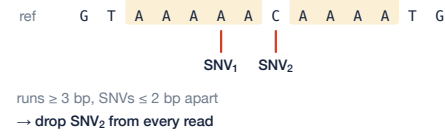
observation quality entering the edge weight

class	source	weight
SNV	aligned base, Phred Q	1 if both bases $\geq$ Q12, else 0.1
indel	CIGAR I/D at the site	1; 0.1 if in a tandem repeat
SV	caller read list	1 for ALT, 0.1 for REF
5mC	modcall read lists	1, strand matched

d Tandem-repeat indel flag

G CA CA CA CA CA T +CA / -CA inside the repeat
 → the indel stays in the graph but votes with weight 0.1

b Homopolymer SNV filter (nanopore)



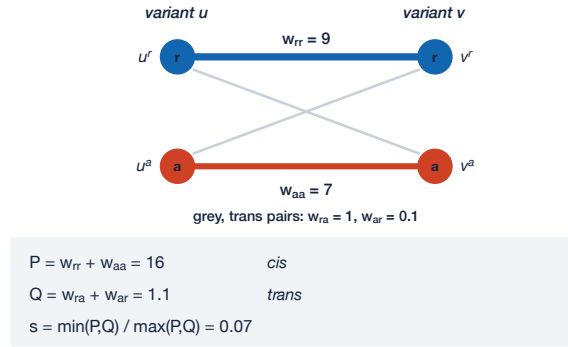
c Overlapping-alignment filter



order of operations extract → b homopolymer → c overlap → d repeat flag → Fig. 4 copy number → graph

Supplementary Fig. 1: Read-level allele observations and pre-graph filters. **a**, For every alignment, each overlapped variant yields an allele observation with an associated quality: SNVs from the aligned base (Phred quality), indels from CIGAR operations starting at the site, SVs from CIGAR operations of matching length or from the caller's read list, and 5mC from the `modcall` read lists. The table summarizes how each class enters the edge weights. **b**, Nanopore-specific homopolymer filter: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV is removed. **c**, Overlapping-alignment filter for supplementary alignments of one read. **d**, Indels in tandem repeats are flagged and vote with weight 0.1. The strip at the bottom gives the order in which the filters are applied; the copy-number-aware filter of Supplementary Fig. 4 acts last, immediately before graph construction.

a Allele-pair weights between two variants



each read adds 1 (both bases $\geq Q_{12}$) or 0.1

d Exported graph (DOT)

```
1001.1 -> 1543.1 [label=16.0]
1001.2 -> 1543.2 [label=1.1]
position.allele -> position.allele, label = vote weight
```

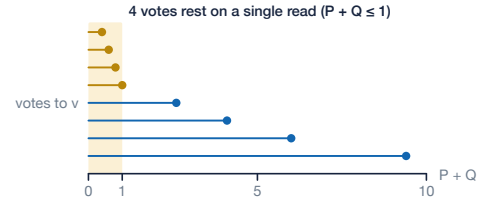
b Vote rules for the pair (u, v)

condition on s	vote for v
$s > 0.7$	no vote
$s > 0.3$, SNV-5mC pair	no vote
$0.1 < s \leq 0.7$	vote larger of P, Q; weight 1
$s \leq 0.1$ or one side unsupported	vote; weight 20
u is a tandem-repeat indel	vote; weight 0.1

the vote names a haplotype for v given hap(u)

$P \rightarrow v^r$ on hap(u^r); $Q \rightarrow v^r$ on hap(u^a)

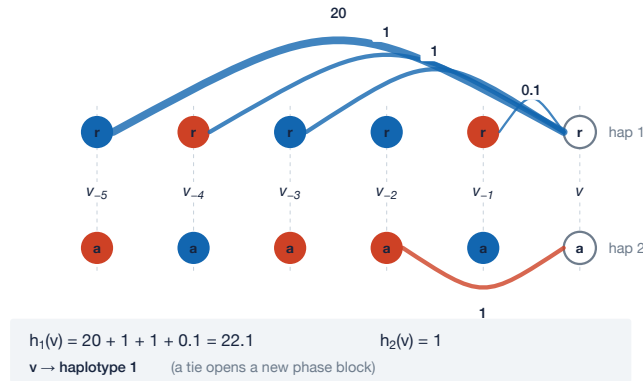
c Single-read guard



> 3 such votes $\rightarrow$ recompute h_1, h_2 from $s < 0.2$, non-indel votes only

Supplementary Fig. 2: Pairwise allele support and vote weighting. **a**, Between two variants u and v the reads accumulate four weights ($w_{rr}, w_{ra}, w_{ar}, w_{aa}$); each read adds 1 when both bases have quality $\geq Q_{12}$ and 0.1 otherwise. P and Q are the supports for the two possible pairings and s their similarity. **b**, Decision rules that turn (P, Q, s) into a weighted haplotype vote for v . **c**, Guard against a single erroneous ultra-long read: among the votes received by v , those whose pair is supported by a single read ($P + Q \leq 1$, amber) are counted, and if more than three such votes exist h_1 and h_2 are recomputed from high-consistency ($s < 0.2$), non-indel votes only. **d**, Format of the exported DOT graph consumed by the **gnn** module.

a Weighted votes from up to $k = 35$ upstream variants

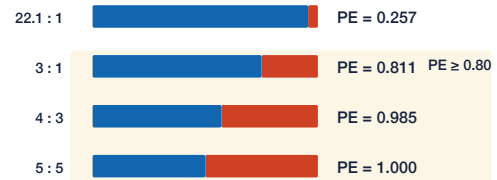


c Block formation



b Phasing entropy

$$PE(v) = -\sum p_i \log_2 p_i, p_i = h_i / (h_1 + h_2)$$



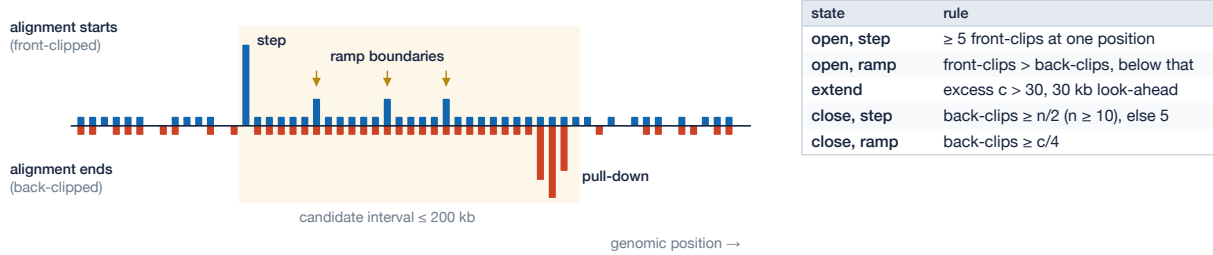
$PE \geq 0.80 \rightarrow$ a GNN window is opened (Fig. 6)

written to VCF as INFO/PE, INFO/H1, INFO/H2

● allele on haplotype 1 ● allele on haplotype 2

Supplementary Fig. 3: Multi-neighbour voting and phasing entropy. **a**, Variant v receives weighted votes from up to $k = 35$ upstream variants within 300 kb; line thickness encodes vote weight (20, 1 or 0.1). The haplotype with the larger total wins; a tie opens a new phase block. **b**, Phasing entropy of the vote totals for four examples; values ≥ 0.80 (dashed box) trigger a GNN window. **c**, Block formation. Variants are visited in coordinate order and each assigned variant casts votes to its next k neighbours within 300 kb; a variant that receives no informative vote, or a tie, starts a new block (amber). Blocks holding a single variant are dropped (grey). Haplotype labels are propagated along the block to give the phased genotypes 0|1 and 1|0, and the phase set is the position of the first variant of the block.

a Two-state detection of copy-number-altered intervals

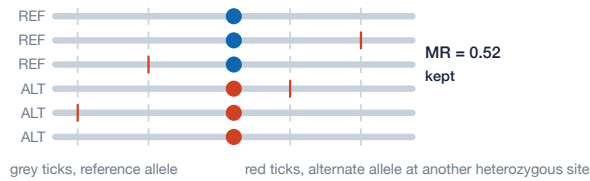


b Mismatch load separates true heterozygotes from paralogues

R_m = alternate alleles carried by one read inside the interval

$$MR = \frac{\text{mean } R_m(\text{ALT})}{\text{mean } R_m(\text{REF}) + \text{mean } R_m(\text{ALT})} \geq 0.7 \rightarrow \text{remove the site from every read}$$

true heterozygous site



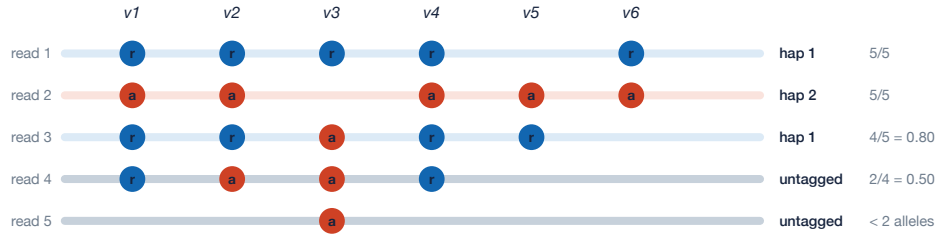
parologue-specific difference



Supplementary Fig. 4: Copy-number-aware filtering of unreliable heterozygous sites. a,

Per-position counts of alignments starting (front-clipped, above the axis) and ending (back-clipped, below) are scanned by a two-state detector. A *step* boundary (≥ 5 front-clips at one position) marks a copy-number transition; a *ramp* boundary (front-clips exceeding back-clips without reaching that threshold) marks the nested breakpoints left by breakage–fusion–bridge cycles. Both states carry a running excess c of front- over back-clips and extend the candidate end by a 30 kb look-ahead while $c > 30$; a step closes at an adaptive pull-down threshold ($n/2$ for an opening front-clip count $n \geq 10$, else 5) or when c decays to its floor, a ramp closes when back-clips reach $c/4$. Candidates longer than 200 kb are abandoned. **b**, Inside an interval, each read is scored by its mismatch load R_m — the number of heterozygous sites in the interval at which it carries the alternate allele — and each heterozygous site by $MR = \bar{R}_m^{\text{ALT}} / (\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}})$. At a genuine heterozygous site both allele populations are ordinary reads of the locus and $MR \approx 0.5$; at a parologue-specific difference the alternate allele is carried by reads that originate elsewhere and mismatch the reference at many other sites in the same interval, so $MR \rightarrow 1$. Sites with $MR \geq 0.7$ are removed from all reads. Sites at which either mean is zero are never removed. [Add a panel **c**: the TP/FP histogram of MR on HG002, with the 0.7 threshold marked. This figure already exists in the group’s algorithm slides and shows the separation directly; regenerate it with the submitted version on the Clair3 calls used here, since the slide version predates it. Without it the threshold is asserted rather than shown.]

a Step 1 — assign each read to a haplotype



tagged if $\max(n_1, n_2) / (n_1 + n_2) > 0.65$ and ≥ 2 informative alleles
allele weights: SNV: 1, SV: 0.1, indel: 0.1, 5mC: 0

b Step 2 — re-derive each variant's phase from tagged reads

variant v_3

$n_1 = 1, n_2 = 2$

n_1 : hap-1 REF + hap-2 ALT reads

n_2 : hap-1 ALT + hap-2 REF reads

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2} = \frac{2}{3} = 0.67$$

$\leq 0.75 \rightarrow \text{unphased, GT 0/1}$

variant v_1

$n_1 = 3, n_2 = 0$

n_1 : hap-1 REF + hap-2 ALT reads

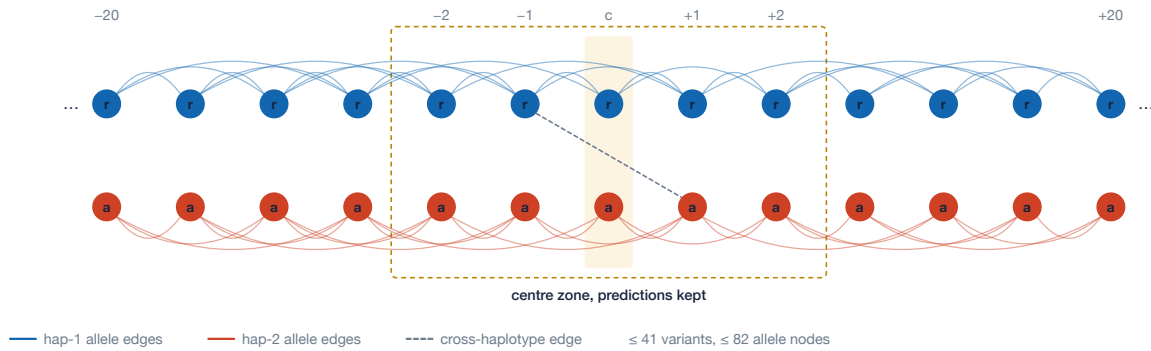
n_2 : hap-1 ALT + hap-2 REF reads

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2} = \frac{3}{3} = 1.00$$

$> 0.75 \rightarrow \text{phased, GT 0/1}$

Supplementary Fig. 5: Read-based correction. a, Each read is assigned to a haplotype by majority over its phased SNV and SV alleles (indels count 0.1, 5mC 0); reads with majority ≤ 0.65 or fewer than two informative alleles remain untagged. b, Each variant's phase is re-derived from the tagged reads and retained only if the consistent fraction ρ exceeds 0.75; otherwise the variant is unphased. Only variants supported by both read populations keep a phase, so assignments driven by a few chimeric or mis-mapped reads are removed. The two thresholds are `-readConfidence 0.65` and `-snpConfidence 0.75`.

a A window is opened around every phased variant with PE ≥ 0.8



b Bridge-vertex feature



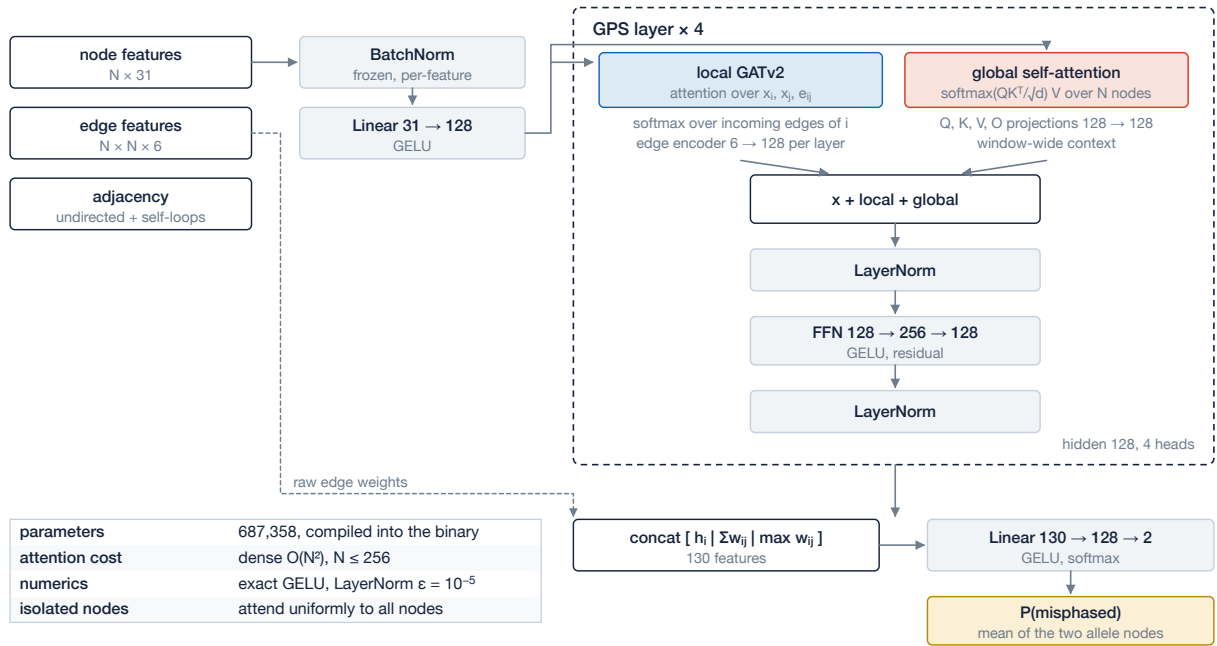
removing this variant disconnects the window
 $\rightarrow \text{is_bridge} = 1$

c Centre-zone mask

$$r = |\text{pos} - \text{pos}_c| / \text{max offset in the window}$$

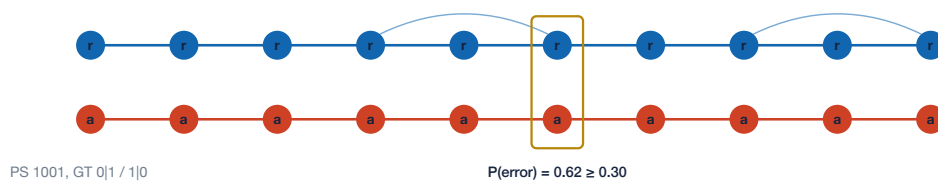
kept if $r \leq \min(1, 10 / \text{round}(20 \cdot r_{\text{max}}))$

Supplementary Fig. 6: Window construction for the graph neural network. a, A window of ± 20 variants (two allele nodes each) is opened around every phased variant with PE ≥ 0.8 ; edges are the DOT edges with both endpoints in the window (blue and red arcs) plus self-loops; a dashed grey edge marks a cross-haplotype link. Predictions are read only in the central zone. b, Bridge-vertex feature: a variant whose removal disconnects the window graph. c, Definition of the central zone. A window holds at most 41 variants and 82 allele nodes, and windows exceeding 256 nodes are skipped; SNV, indel, SV and 5mC records are merged and sorted by position before nodes are created. Edges are the DOT edges with both endpoints in the window, symmetrized, plus one self-loop per node whose features are the mean over that node's incident edges. The predictions of the two allele nodes of a variant are averaged, and a variant covered by several windows receives the mean over those windows.

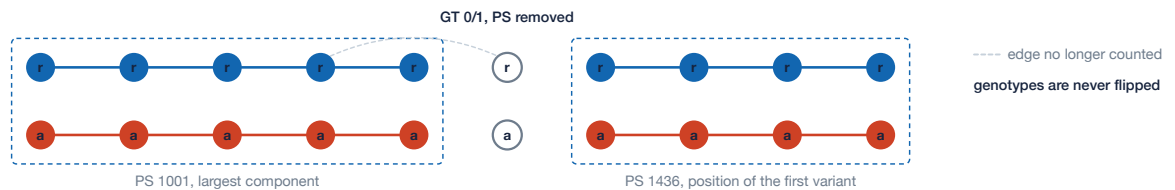


Supplementary Fig. 7: Graph neural network architecture. Node features are standardized and projected to 128 dimensions. Each of four GPS layers combines a local GATv2 branch over the window graph (edge features enter through a per-layer edge encoder) with a global self-attention branch over all nodes, fused additively into the residual stream and followed by layer normalization and a feed-forward block. The final embedding is concatenated with the sum and maximum of incoming raw edge weights and classified into phased-correctly versus misphased; the probabilities of the two allele nodes of a variant are averaged.

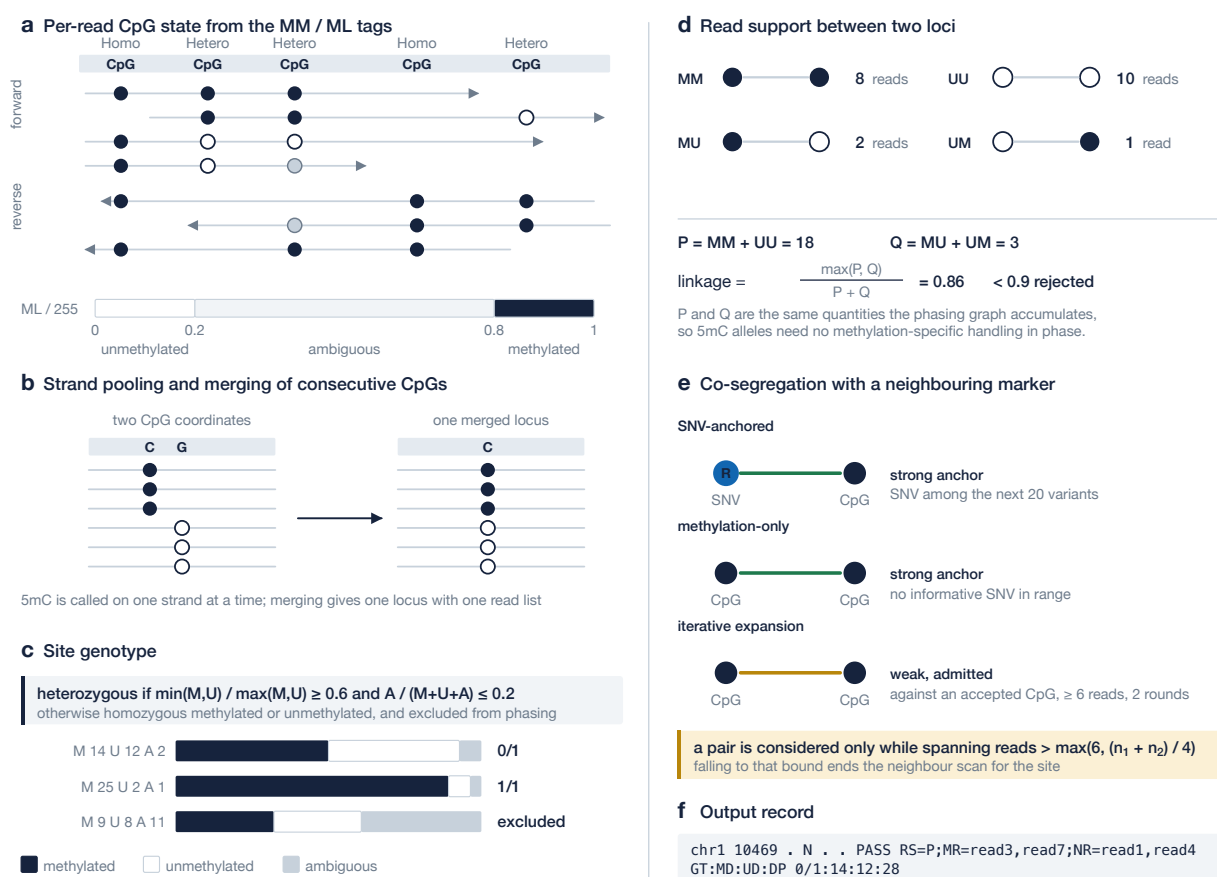
a Before correction — one phase set



b After correction — variant unphased, phase set split

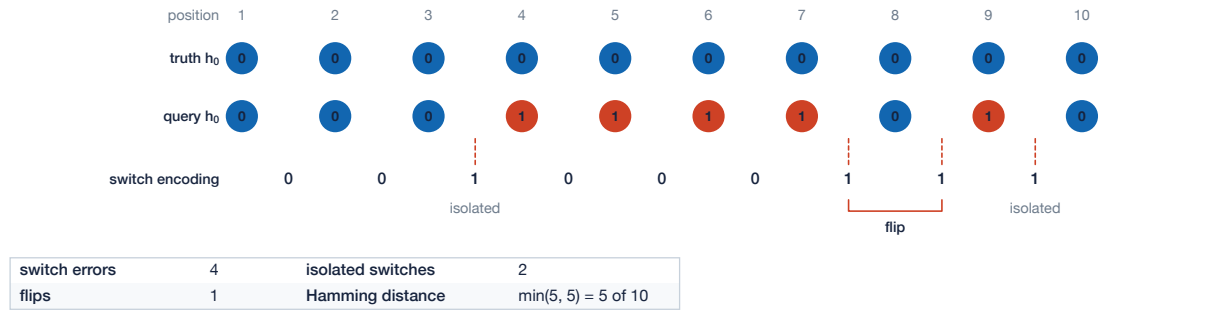


Supplementary Fig. 8: Unphasing and phase-set splitting. **a**, A variant with predicted error probability ≥ 0.30 (amber) inside a phase set. **b**, After correction the variant is written as an unphased genotype without PS; connectivity of the remaining variants is recomputed over the surviving DOT edges, the largest component keeps the original PS and every other component receives a new PS equal to the position of its first variant. Genotypes are never flipped.



Supplementary Fig. 9: Allele-specific methylation calling with modcall. **a**, Per-read CpG state is read from the MM/ML tags on both strands: methylated at $ML/255 \geq 0.8$, unmethylated at ≤ 0.2 , ambiguous in between. Only CpG dinucleotides are considered. **b**, 5mC is called on one strand at a time, so reverse-strand calls are mapped to the forward CpG coordinate and runs of consecutive CpGs are merged into a single locus carrying one read list. This removes strand from the problem and lets phase vote on methylation loci with the unmodified graph algorithm. **c**, A site is heterozygous when $\min(M, U) / \max(M, U) \geq 0.6$ and $A / (M + U + A) \leq 0.2$, for methylated, unmethylated and ambiguous read counts M , U and A ; otherwise it is homozygous and excluded. **d**, Balance alone does not imply allele specificity, since stochastic methylation satisfies **c** as well. Read support between two loci is therefore decomposed into the four state pairs, whose concordant and discordant totals are exactly the quantities P and Q that the phasing graph accumulates (`VariantEdge::findNumberOfRead`); a pair is linked when $\max(P, Q) / (P + Q) \geq 0.9$. The example shown falls below the threshold and is rejected. **e**, Linking runs in three modes: against an SNV among the next 20 variants, against another candidate CpG where no informative SNV is in range, and an iterative expansion that admits weak candidates against an already accepted CpG. In every mode a pair is considered only while the reads spanning it exceed $\max(6, (n_1 + n_2) / 4)$, and falling to that bound ends the neighbour scan for that site rather than skipping one pair. **f**, Output record; the MR and NR read names attach the 5mC alleles to reads in phase. [Two real-data panels would close the remaining gaps, and both exist in the group's thesis material. (i) Per-site ML probability at heterozygous versus homozygous CpGs with the 0.8 and 0.2 lines marked, showing that it is the homozygous sites that spread across the ambiguous band; the original was computed on 48 sites in the imprinted *SNRPN* locus, too small and too special a sample for a Nature Methods supplement, so recompute genome-wide on HG002. (ii) Distribution of the linkage ratio in **d** stratified by the true class of the site pair (heterozygous–heterozygous, homozygous–homozygous, heterozygous–homozygous). This is the strongest justification available for the 0.9 threshold: in the original the three classes separate at roughly 0.65–1, 0.5–0.75 and 0.5–0.7, showing that the cut removes homozygous sites misclassified as allele-specific rather than being an arbitrary round number. Regenerate both with the submitted version.]

a Switch errors, flips and Hamming distance within one block



b Block statistics



Supplementary Fig. 10: Phasing evaluation metrics. **a**, Within one intersected block, the query haplotype string is compared with the truth in switch encoding; the number of differing positions is the switch-error count, adjacent switch pairs are counted as flips, and the Hamming distance is the minimum over the two orientations of misassigned variants. **b**, Block intersection and the block-level statistics. Only variants that are heterozygous in both files and phased in both are compared (dark bars); blocks left with a single variant are ignored (outline). Phased fraction, block N50 and switch error rate are computed over the intersected blocks as implemented in `longphase compare`.

Placeholder — panels to be regenerated with the submitted version.

Supplementary Fig. 11: Calibration of the voting and read-filtering constants. Each panel is a sweep on HG002 at 10–60 $\times$; the value adopted in LongPhase 2 is marked. **a**, Distribution of pair ambiguity $s = \min(P, Q) / \max(P, Q)$ over all assessed pairs and over the subset that participate in a switch error, showing that both mass and residual error concentrate below $s = 0.1$ and motivating $\alpha = 0.1$ as the boundary of the up-weighted class. **b**, Switch errors and block N50 as a function of the no-vote threshold β . Switch errors fall monotonically as β decreases, so switch error alone does not identify an optimum; $\beta = 0.7$ is the smallest value at which block N50 is not yet degraded, and is adopted on that basis. **c**, Switch errors as a function of the up-weight w applied when $s < \alpha$, at each coverage; the reduction saturates by $w \approx 20$. **d**, Distribution of base quality over all bases and over bases carrying a sequencing error, showing errors concentrated below Q10 and motivating $\theta = 12$; and switch errors as a function of θ . **e**, Switch errors over the grid of single-read-guard parameters N (number of single-read votes that triggers the guard) and μ (ambiguity cut applied to the surviving votes), and switch errors with and without the guard at each coverage. [All five panels exist as figures in the group's thesis material (Po-Hui Tzeng, 2026, Figs. 3.7, 3.10–3.12, 3.14, 3.17, 3.18, 3.21) and should be regenerated with the submitted version on the Clair3 calls and coverages used here, then redrawn to a single consistent style. Two caveats to carry over honestly: in **b** the switch-error curve does not have an interior optimum, so the choice of β rests on the N50 trade-off and must be presented that way; and in **e** the guard slightly *increased* switch errors at 10 $\times$ in the original sweep, which should be shown rather than cropped.]

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